

CHAPTER 1 RESULTS AND DISCUSSION

This chapter presents the experimental results of the proposed FeatureSAGE framework and compares its performance with the AGE and SAGE baselines on few-shot image generation tasks. The experiments are conducted on the FUNIT Animal Faces and 102 Flowers datasets under a one-shot setting, where a single reference image is used for each target class during evaluation. All reported quantitative results are evaluated at 1024×1024 resolution. Generated-image quality is measured using Fréchet Inception Distance (FID), and perceptual variation is measured using Learned Perceptual Image Patch Similarity (LPIPS). The comparison isolates the effect of replacing the pSp-centered SAGE inversion stage with the StyleFeatureEditor inverter and Feature Editor while keeping the same baseline task and evaluation setting.

1.1 Quantitative Results

1.1.1 Quantitative Comparison Between FeatureSAGE and Baselines

Resolution	Model	Flowers		Animal Faces	
		FID ↓	LPIPS ↑	FID ↓	LPIPS ↑
1024	AGE (Baseline)	61.57	0.4108	62.13	0.5033
1024	SAGE (Baseline)	54.17	0.4528	60.92	0.4867
1024	FeatureSAGE (Ours)	45.60	0.4209	49.60	0.4811

Table 1.1 Quantitative comparison of AGE, SAGE, and the proposed FeatureSAGE framework across the 102 Flowers and Animal Faces datasets at 1024×1024 resolution. Lower FID indicates better distributional alignment with real images. Higher LPIPS is interpreted here as greater perceptual variation among generated outputs.

1.2 Analysis of Quantitative Results

This section analyzes the quantitative trends in Table 1.1. The comparison focuses on how FeatureSAGE performs against AGE and SAGE across the 102 Flowers and Animal Faces datasets at 1024×1024 resolution using FID and LPIPS.

1.2.1 102 Flowers

On the 102 Flowers dataset, FeatureSAGE obtains an FID of 45.60, corresponding to a 15.8% improvement over SAGE (54.17) and a 25.9% improvement over AGE (61.57). This FID reduction indicates stronger distributional alignment with real flower images in the Inception feature space. For perceptual variation, FeatureSAGE achieves an LPIPS of 0.4209, compared with 0.4528 for SAGE and 0.4108 for AGE. This represents a 7.0% decrease relative to SAGE and a 2.5% increase relative to AGE. The LPIPS values across the three methods range from 0.4108 to 0.4528, showing that perceptual variation remains within a narrow range while FeatureSAGE achieves the best FID.

1.2.2 Animal Faces

On the Animal Faces dataset, FeatureSAGE achieves an FID of 49.60, outperforming SAGE (60.92) by 18.6% and AGE (62.13) by 20.2%. This reduction indicates improved distributional alignment with the real animal-face data. For LPIPS, FeatureSAGE obtains 0.4811, compared with 0.4867 for SAGE and 0.5033 for AGE. This is a 1.2% decrease relative to SAGE and a 4.4% decrease relative to AGE. The LPIPS values remain close across the three methods, ranging from 0.4811 to 0.5033, which indicates that FeatureSAGE improves FID while maintaining comparable perceptual variation.

1.3 Qualitative Evaluation

1.3.1 Visual Comparison Across Models



Figure 1.1 Comparison across AGE, SAGE, and the proposed FeatureSAGE model under the one-shot setting, where a single reference image is used to condition the model during inference.



Figure 1.2 Comparison across AGE, SAGE, and the proposed FeatureSAGE model under the one-shot setting with a second flower input.



Figure 1.3 Comparison across AGE, SAGE, and the proposed FeatureSAGE model on the Animal Faces dataset under the one-shot setting. Each model generates two outputs per input image to demonstrate edit diversity and semantic consistency.



Figure 1.4 Comparison across AGE, SAGE, and the proposed FeatureSAGE model under the one-shot setting with a second dog input image. Each model generates two outputs per input to illustrate diversity and semantic consistency.

1.3.2 Visual Observations

Flowers

Figures 1.1 and 1.2 compare generated flower images from AGE, SAGE, and FeatureSAGE. The visual analysis focuses on color preservation, edge sharpness, contrast, and semantic consistency. In Figure 1.1, FeatureSAGE retains the pink petal color of the input image and produces sharper petal boundaries than the baseline outputs. The generated image also shows stronger contrast, which contributes to a more defined flower structure. However, the model does not fully preserve the combined red and yellow colors of the pistil from the input image, indicating that fine internal flower details may still be difficult to reconstruct consistently.

Figure 1.2 further shows that FeatureSAGE preserves petal-edge clarity and produces visually coherent flower outputs. One generated output presents a slight tilt, demonstrating that the model can introduce appearance variation while retaining the flower category. Compared with AGE, which appears paler in some regions, and SAGE, which improves contrast but remains less sharp in the shown examples, FeatureSAGE produces clearer and more realistic flower structures in the qualitative comparisons.

Animal Faces

Figure 1.3 compares AGE, SAGE, and FeatureSAGE on an Animal Faces input. All models preserve the basic color and fur texture of the input, but the outputs differ in structural consistency and local realism. AGE and SAGE show more visible variation in orientation between their two generated outputs. FeatureSAGE produces more consistent pose and structure while maintaining coherent fur shading and blending. This supports the quantitative observation that FeatureSAGE improves fidelity while preserving usable perceptual variation.

Figure 1.4 shows similar behavior for the second dog input. FeatureSAGE maintains more consistent eye shape and facial structure than the baselines in the displayed results. Its outputs also show clearer fur texture and stronger contrast. AGE produces greater pose diversity in this example, but the added variation is accompanied by reduced visual clarity. These observations are consistent with the LPIPS results, where FeatureSAGE remains close to the baselines in perceptual variation while achieving the lowest FID on the Animal Faces dataset.

1.4 Failure Cases and Limitations

During experimentation and evaluation of the FeatureSAGE framework, two main limitations were observed:

1. **Dependence on a pretrained StyleGAN2 generator.** FeatureSAGE inherits the coverage and biases of the pretrained StyleGAN2 generator. If the generator does not adequately represent certain categories or visual attributes, those gaps may appear in the generated outputs. This constraint limits the framework to domains where an appropriate pretrained StyleGAN2 model is available and sufficiently aligned with the target data.
2. **Computational overhead.** Integrating the StyleFeatureEditor inverter and Feature Editor increases the computational cost relative to the baseline SAGE setup. These components improve reconstruction and feature-level editing, but they require additional generator passes and feature-tensor operations. As a result, the framework requires

more GPU memory and can slow inference or training, especially when scaling to larger datasets or time-sensitive applications.

These limitations indicate that the reported quality improvements involve a trade-off between image fidelity and computational efficiency. They also motivate future work on lighter inversion modules, more efficient feature editing, and evaluation with generator backbones beyond StyleGAN2.

1.5 Chapter Summary

The results show that FeatureSAGE improves distributional image quality over both baselines on the two evaluated datasets. On 102 Flowers, FeatureSAGE achieves an FID of 45.60, improving over SAGE by 15.8% and over AGE by 25.9%. On Animal Faces, it achieves an FID of 49.60, improving over SAGE by 18.6% and over AGE by 20.2%. LPIPS remains comparable to the baselines, with 0.4209 on Flowers and 0.4811 on Animal Faces, indicating that the FID gains do not remove perceptual variation among generated outputs. Qualitative comparisons further show sharper flower structures, stronger petal clarity, and more coherent animal-face texture and shading in the FeatureSAGE outputs, while the limitations indicate increased computational overhead from the added inverter and Feature Editor components.